

# **50.007 Machine Learning**

## **Lecture 19**

# **Reinforcement Learning**

# Markov Decision Process (MDP)

An MDP is defined by:

- A **set of states**  $s \in S$
- A **set of actions**  $a \in A$
- A **transition function**  $T(s, a, s')$ 
  - Probability that  $a$  from  $s$  leads to  $s'$ , i.e.,  $P(s' | s, a)$
  - Also called the model or the dynamics
- A **reward function**  $R(s, a, s')$ 
  - Sometimes just  $R(s)$  or  $R(s')$
- A **start state**
- Maybe a **terminal state**

|        |        |        |        |
|--------|--------|--------|--------|
| 0.64 ▶ | 0.74 ▶ | 0.85 ▶ | 1.00   |
| 0.57 ▲ |        | 0.57 ▲ | -1.00  |
| 0.49 ▲ | 0.43 ◀ | 0.48 ▲ | 0.28 ◀ |

How do we  
learn the policy?

|   |   |   |      |
|---|---|---|------|
| → | → | → | +1.0 |
| ↑ |   | ↑ | -1.0 |
| ↑ | ← | ↑ | ←    |

# Markov Decision Process (MDP)

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# Markov Decision Process (MDP)

An MDP is defined by:

- A set of states  $s \in S$
- A set of actions  $a \in A$

How do we deal with the unknown transition probabilities and the reward functions?

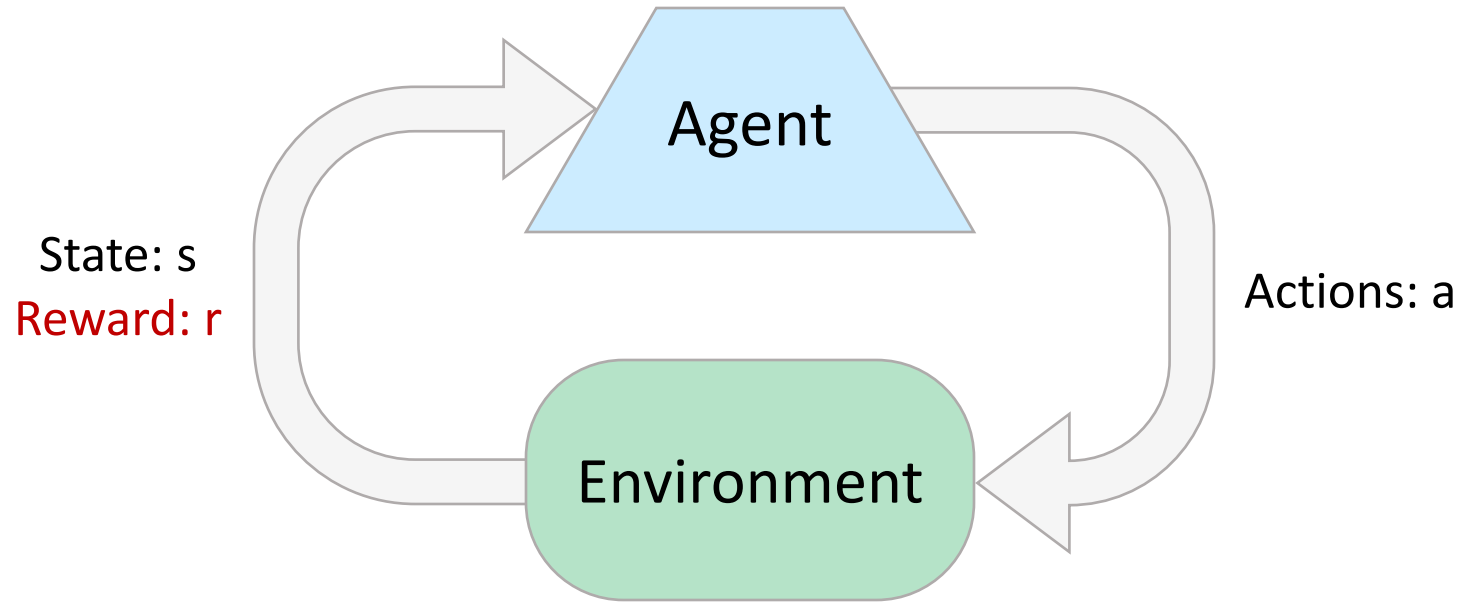
• Sometimes just  $R(s)$  or  $R(s, a)$

- A start state
- Maybe a terminal state

How do we learn the policy?



# Reinforcement Learning



- Basic idea:
  - Receive feedback in the form of **rewards**
  - Agent's utility is defined by the reward function
  - Must (learn to) act so as to **maximize expected rewards**
  - All learning is based on observed samples of outcomes!

# Reinforcement Learning



How do we deal with the unknown transition probabilities and the reward functions?



Imagine the robot is moving around...

$$T(s'|s, a) = \frac{\text{Count}(s, a, s')}{\text{Count}(s, a)}$$

$$R(s, a, s') = \frac{\sum_t R_t(s, a, s')}{\text{Count}(s, a, s')}$$

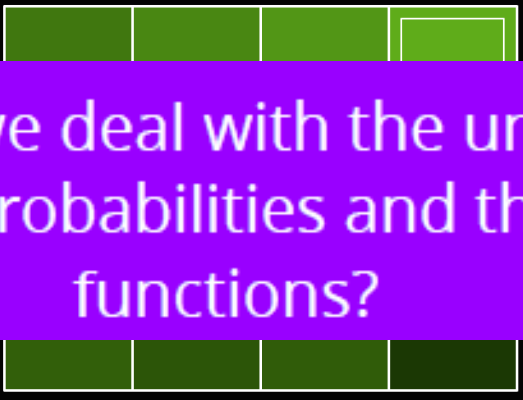
**Model-based Approach**

Estimating the model parameters first

eg. Robot went North  
to x and ended up at  
2x  $\Rightarrow$  Count of robot @ states s taken & arriving at s'

$$\frac{\text{Count of robot @ states s taken & arriving at s'}}{\text{Count of robot @ states s taken}} = \frac{2}{10} = 0.2$$

# Reinforcement Learning



How do we deal with the unknown transition probabilities and the reward functions?

Imagine the robot is moving around...

$$T(s'|s, a) = \frac{\text{Count}(s, a, s')}{\text{Count}(s, a)} \quad R(s, a, s') = \frac{\sum_t R_t(s, a, s')}{\text{Count}(s, a, s')}$$

**Model-free** Approach

Does not explicitly estimate parameters

# Example: A Simple Game

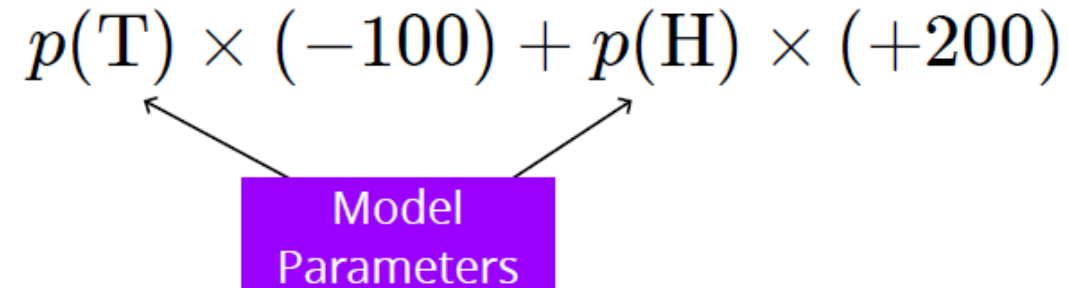
Flipping a coin, if it is head (H), you receive \$200; otherwise you pay \$100. Would you like to play the game?

# Model-Based Approach

Flipping a coin, if it is head (H), you receive \$200; otherwise you pay \$100. Would you like to play the game?

$$p(T) \times (-100) + p(H) \times (+200)$$

Model  
Parameters



# Model-Based Approach

H

H

T

T

H

T

T

T

T

T

T

$$p(H) = 0.3$$

$$p(T) = 0.7$$

$$\begin{aligned} & p(T) \times (-100) + p(H) \times (+200) \\ &= -70 + 60 = -10 \end{aligned}$$

# Model-Based Approach

## Drawbacks

- Requires random exploration
- Cannot explore all states in complex environments
- Learned model would be quite large as we need to store all the states and possible transitions.

Can we learn  
a policy without  
learning a full  
model?

# Simple Game: Model-Free Approach

|   | Outcome | Gain/Loss | Average Return |
|---|---------|-----------|----------------|
| 1 | H       | +200      | +200           |

Estimate  
expectation  
directly from  
samples



# Model-Free Approach

|   | Outcome | Gain/Loss | Average Return |
|---|---------|-----------|----------------|
| 1 | H       | +200      | +200           |
| 2 | H       | +200      | +200           |

Estimate  
expectation  
directly from  
samples

# Model-Free Approach

|   | Outcome | Gain/Loss | Average Return |
|---|---------|-----------|----------------|
| 1 | H       | +200      | +200           |
| 2 | H       | +200      | +200           |
| 3 | T       | -100      | +100           |

Estimate  
expectation  
directly from  
samples

# Model-Free Approach

|    | Outcome | Gain/Loss | Average Return |
|----|---------|-----------|----------------|
| 1  | H       | +200      | +200           |
| 2  | H       | +200      | +200           |
| 3  | T       | -100      | +100           |
|    | T       | -100      |                |
|    | H       | +200      |                |
|    | T       | -100      |                |
|    | T       | -100      |                |
|    | T       | -100      |                |
|    | T       | -100      |                |
| 9  | T       | -100      |                |
| 10 | T       | -100      | -10            |

Estimate  
expectation  
directly from  
samples

# Model-Free Approach

|         | Outcome | Gain/Loss | Average Return |
|---------|---------|-----------|----------------|
| 1       | H       | +200      | +200           |
| 2       | H       | +200      | +200           |
| 3       | T       | -100      | +100           |
|         | ⋮       | ⋮         | ⋮              |
| $k - 1$ | T       | -100      | $v$            |

Estimate  
expectation  
directly from  
samples

# Model-Free Approach

|         | Outcome  | Gain/Loss | Average Return |
|---------|----------|-----------|----------------|
| 1       | H        | +200      | +200           |
| 2       | H        | +200      | +200           |
| 3       | T        | -100      | +100           |
|         | $\vdots$ | $\vdots$  | $\vdots$       |
| $k - 1$ | T        | -100      | $v$            |
| $k$     | H        | +200      | ??             |

Estimate  
expectation  
directly from  
samples

# Model-Free Approach

|         | Outcome | Gain/Loss | Average Return                      |
|---------|---------|-----------|-------------------------------------|
| 1       | H       | +200      | +200                                |
| 2       | H       | +200      | +200                                |
| 3       | T       | -100      | +100                                |
|         | ⋮       | ⋮         | ⋮                                   |
| $k - 1$ | T       | -100      | $v$                                 |
| $k$     | H       | +200      | $\frac{v \times (k-1) + (+200)}{k}$ |

Estimate  
expectation  
directly from  
samples

# Model-Free Approach

$$\frac{v(k-1)}{k} + \frac{(+200)}{k}$$

Total reward @  $k-1$       Immediate reward @  $k$

$$v \leftarrow \frac{v \times (k-1) + (+200)}{k}$$

Estimate expectation directly from samples

$$\frac{1}{k} (v(k-1) + (+200))$$

$$v \leftarrow v + \frac{1}{k} ((+200) - v)$$

$$\frac{1}{k} (vk - v + (+200))$$

Current estimate of Q-value

From a newly collected sample

$$\frac{1}{k} (vk + (+200) - v)$$

$$\frac{vk}{k} + \frac{(+200)}{k} - \frac{v}{k} = v + \frac{1}{k} [(+200) - v]$$

# Model-Free Approach

$$v \leftarrow \frac{v \times (k-1) + (+200)}{k}$$

Estimate  
expectation  
directly from  
samples

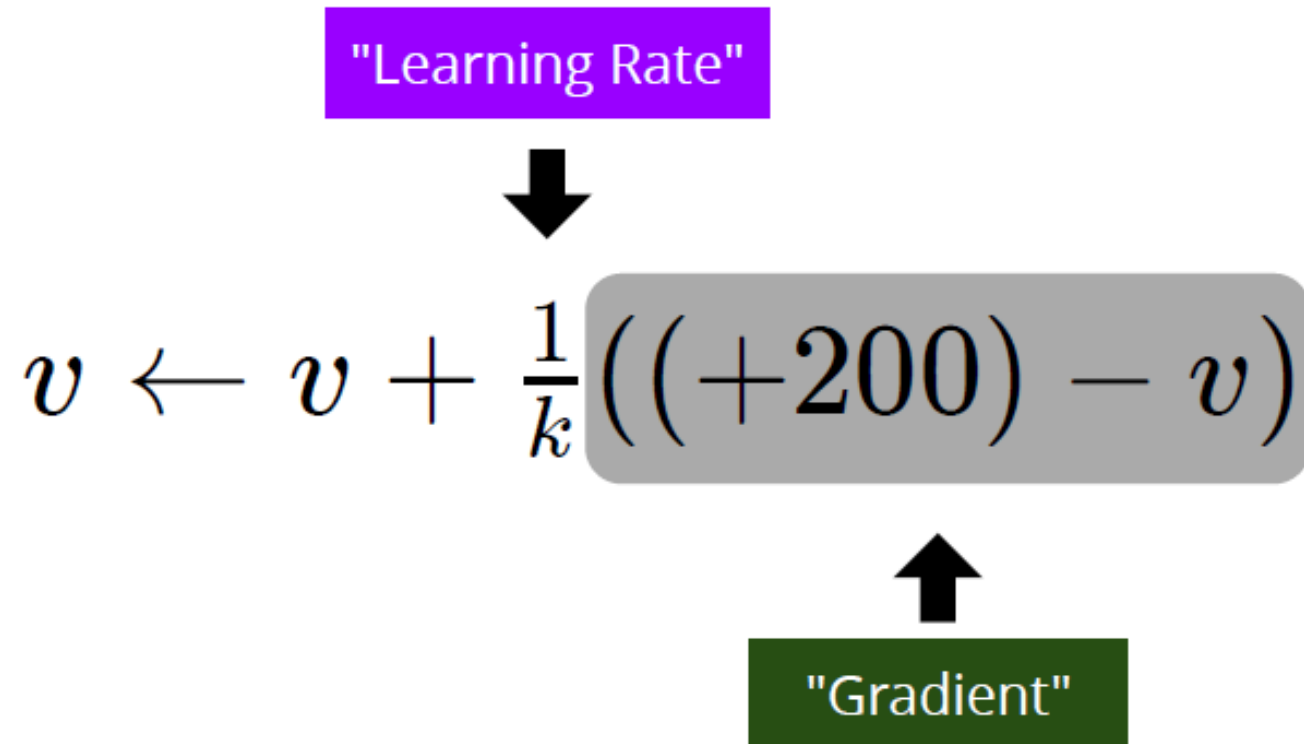
$$v \leftarrow v + \frac{1}{k} ((+200) - v)$$

Current estimate  
of Q-Value

From a newly  
collected sample



# Model-Free Approach



# Updating Q-Values

$$\text{sub } v = Q(s, a)$$

$$\text{sub } (r_0) = R(s, a, s'_k)$$

We have arrived at the following model-free based estimation of the Q-values  
(after taking action a from state s)

$$Q(s, a) \leftarrow Q(s, a) + \frac{1}{k} \left[ R(s, a, s'_k) + \gamma \max_{a'} Q(s'_k, a') - Q(s, a) \right]$$

*Future discounted reward*

$$v \leftarrow v + \frac{1}{k} \left[ (r_0) + \boxed{\phantom{000}} - v \right]$$

# Q-Learning

Collect a sample:  $s, a, s'$  and  $R(s, a, s')$ .



Update Q-values, by incorporating the new sample into a running average over samples:

$$Q(s, a) \leftarrow Q(s, a) + \frac{1}{k} \left[ R(s, a, s'_k) + \gamma \max_{a'} Q(s'_k, a') - Q(s, a) \right]$$

↑

Can be replaced by  $\alpha$

A term introduced like HMU to ensure completeness,  $\therefore$  we need to take into account future discounted rewards.

$$\alpha = \frac{1}{k}$$

# Q-Learning

So when  $k$  is  $\downarrow$  during the  
start  $\alpha \uparrow$  and

Collect a sample:  $s, a, s'$  and  $R(s, a, s')$ .



$(1-\alpha) \downarrow$  so we put less

significance to experience  $= Q(s, a)$

however when  $k \uparrow$ ,  $\alpha \downarrow$

and  $(1-\alpha) \uparrow$ ,

$\therefore$  putting more

Update Q-values, by incorporating

the new sample into a running average over samples:

significance to experience

at later iterations

instead of the current

reward of state

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[ R(s, a, s') + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

$$Q(s, a) \leftarrow (1 - \alpha)Q(s, a) + \alpha \left[ R(s, a, s') + \gamma \max_{a'} Q(s', a') \right]$$

As  $k$  increases, alpha decreases, giving more importance to the old Q-value estimates ( $1 - \alpha$  term).

# Q Learning

Collect a sample:  $s, a, s'$  and  $R(s, a, s')$ .

How shall we  
choose the action?

Update Q-values, by incorporating  
the new sample into a running average over samples:

$$Q(s, a) \leftarrow Q(s, a) + \frac{1}{k} [R(s, a, s') + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

# Exploration vs Exploitation

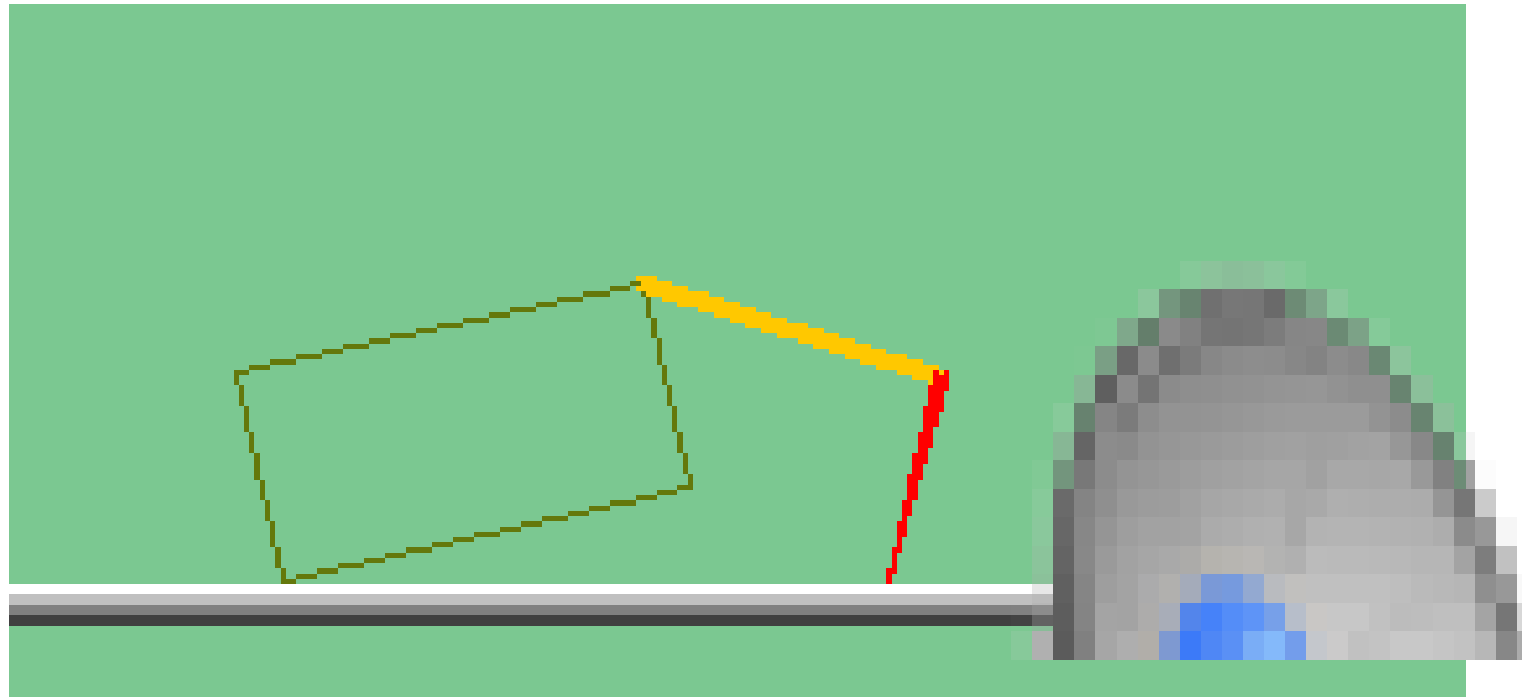
Collect a sample:  $s, a, s'$  and  $R(s, a, s')$ .

Initially, we may randomly pick the action  $a$  (exploration). Later, when we are more confident about the learned Q-values, we may follow the action based on Q-values (exploitation)

→ By reducing  
to the  
epsilon  
value = reduce  
randomness

$$Q(s, a) \leftarrow Q(s, a) + \frac{1}{k} [R(s, a, s') + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

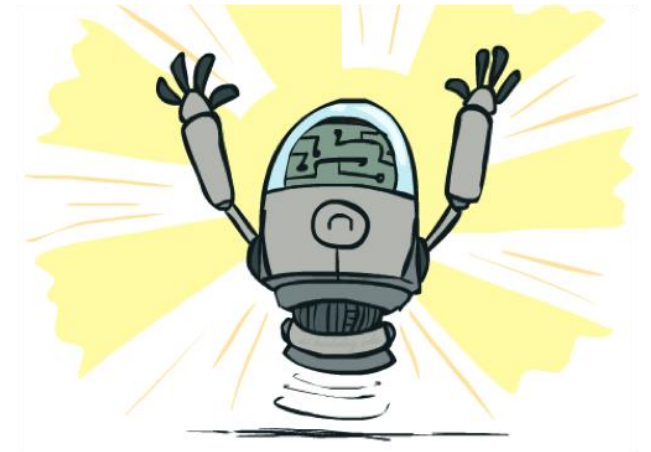
# Q-Learning – Epsilon Greedy - Crawler



Selection a random action with probability epsilon and with 1-epsilon probability follow policy induced by the current Q-values.

# Q-Learning Properties

- **Amazing result:** Q-learning converges to optimal policy -- even if you're acting sub-optimally!
- **Caveats:**
  - You have to explore enough
  - You have to eventually make the learning rate small enough  $\alpha \uparrow = \frac{1}{k} \downarrow$
  - ... but not decrease it too quickly = reduce gradually
  - Basically, in the limit, it doesn't matter how you select actions (!)





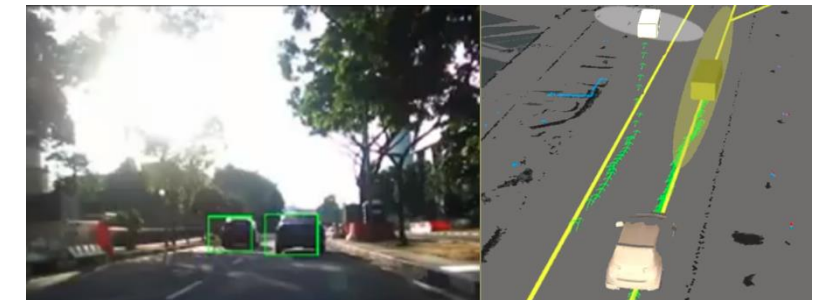
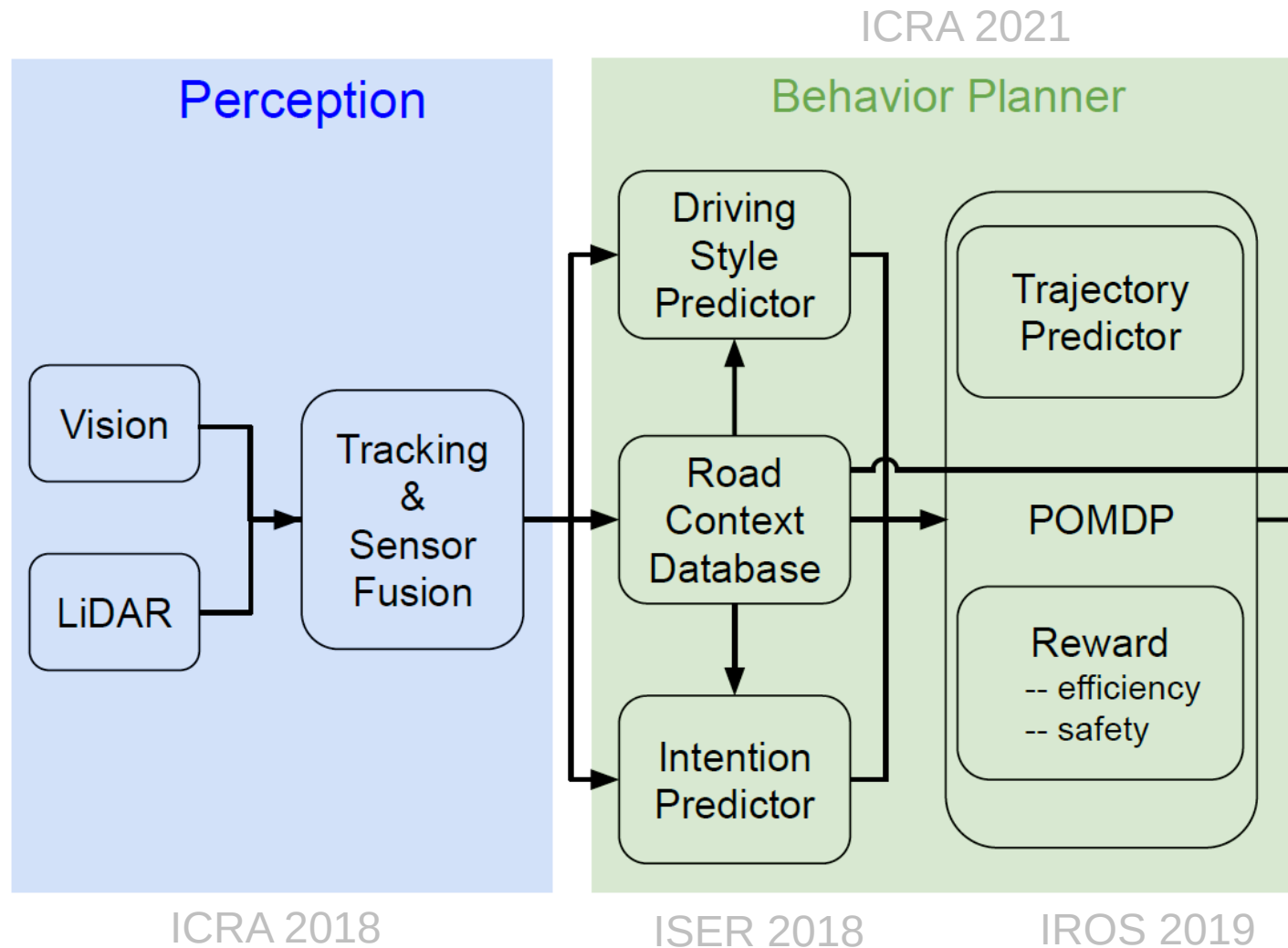
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# **Autonomous Urban Driving**

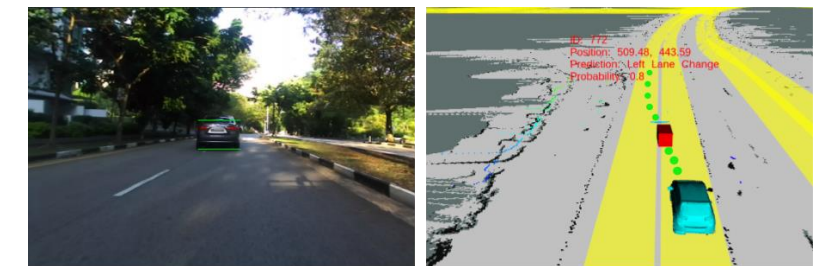
## **[Partially Observable MDP]**

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# Context, Intent and Driving Style aware Planner



**Driving style prediction**



**Driver intent prediction**

# Planning and Decision Making

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We propose a **partially observable Markov decision process (POMDP)** model for **planning and high-level decision making**.

**State (S):** road context (fully observable)  
pose of each vehicle (fully observable)  
acceleration of each vehicle (fully observable)  
speed of each vehicle (fully observable)  
driver intention (partially observable)  
driving style (partially observable)

**Action (A):** lane-keep, left-lane change, right-lane change

**Observation:** road context, poses, acceleration, speeds

**Transition:** trajectory prediction, given the road context, pose, acceleration, speed, predicted intent and predicted driving style

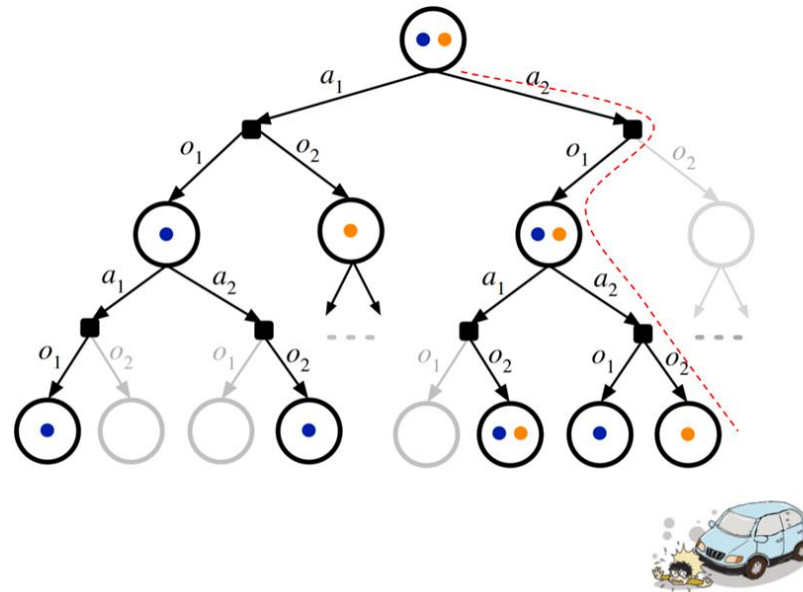
**Reward:**

**Safety:** Penalize collisions

**Efficiency:** Penalize slow speeds (local) and longer paths (global)

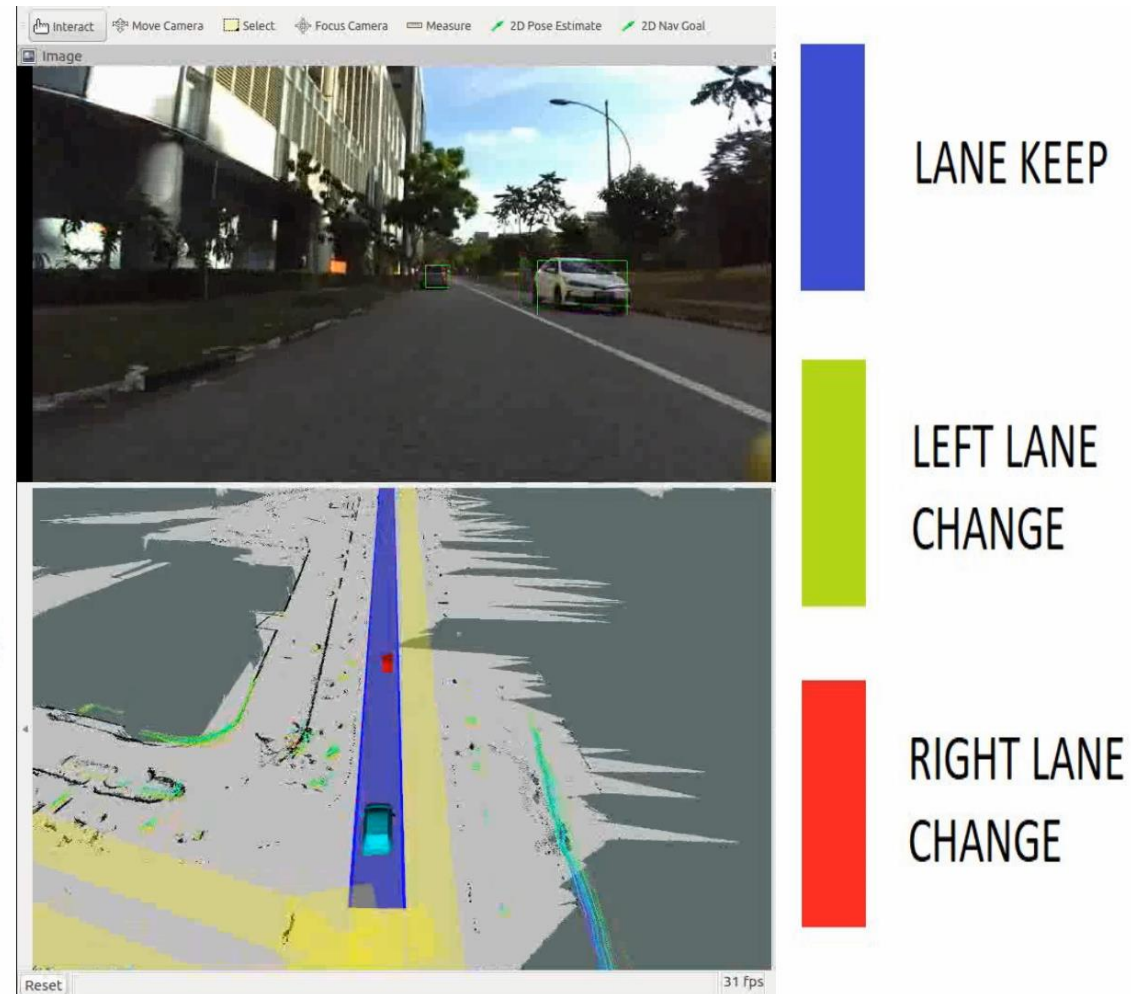
# Context, Intent and Driving Style aware Planner

- The **POMDP** formulation is made **computationally feasible** using the **predicted vehicle intentions, predicted driving styles, trajectories** and **road context**.



- We use **Determinized Sparse Partially Observable Tree (DESPOT)** to solve the POMDP formulation.

# Context and Intent Aware Planning



- **M. Meghjani**, Y. Luo, Q. H. Ho, P. Cai, S. Verma, D. Rus, and D. Hsu, "Context and Intention Aware Planning for Urban Driving", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2019.

# Context, Intent and Driving Style Aware Planning

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## SIMULATION RESULTS

# Context, Intent and Driving Style Aware Planning

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## REAL WORLD RESULTS

# Thank You!!